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EXTRATERRESTRIAL LANDER FOR DATA COLLECTION AND RF
TRANSMISSION

POST-LAUNCH ASSESSMENT REVIEW

NASA STUDENT LAUNCH COMPETITION

2024 - 2025

MAY 19, 2025

Table of Contents

1	Summary	2
1.1	Competition Launch Summary	2
1.2	Vehicle Summary	2
1.3	Payload Summary	2
2	Assessment of Flight	3
2.1	Flight Mission Criticality	3
2.2	Flight Sequence	4
2.3	Flight Performance	4
2.4	Recovery Performance	7
2.5	GPS Data	9
3	Assessment of Vehicle	9
3.1	Vehicle Mission Criticality	9
3.2	Vehicle Design Overview	10
3.3	Vehicle Preparation	10
3.4	Vehicle Reflight Certification	11
3.5	Vehicle Lessons Learned	12
4	Assessment of Payload	13
4.1	Payload Mission Criticality	13
4.2	Design Overview	13
4.3	Payload Retention and Detachment	14
4.4	Mechanism Performance	15
4.5	Payload Reflight Certification	15
4.6	Payload Lessons Learned	16
5	Assessment of Experiment	17
5.1	Experiment Mission Criticality	17
5.2	Software Performance and Data Collection	18
5.3	Protocol and Interference Risks	19
5.4	STEMCRaFT Descent and Landing Analysis	21
6	Project Management	22
6.1	Team Hours Spent	22
6.2	STEM Engagement Summary	22
6.3	Final Budget Summary	23
6.4	Team Experience and Project Lessons Learned	24
A	Vehicle and Payload Reflight Certification	25
B	References	30

1 Summary

1.1 Competition Launch Summary

Table 1: VADL 2025 competition launch summary.

Parameter	Specification
Team Name	Vanderbilt Aerospace Design Laboratory
Vehicle Name	<i>The Bucket</i>
Payload	Extraterrestrial Lander for Data Collection and RF Transmission
Launch Site	Bragg Farms, Toney, Alabama
Target Apogee	4200 ft
Apogee Reached	4428 / 4429 ft (Main / Backup Altimeter)

1.2 Vehicle Summary

The 2025 competition vehicle, “*The Bucket*”, was successfully flown and recovered on May 4, 2025 at Bragg Farms in Toney, AL. Key vehicle parameters are listed in Table 2. The Bucket reached an apogee within 229 ft of the target and safely retained the payload until jettison. Following landing, inspections revealed that no damage was incurred to any vehicle sections.

Table 2: *The Bucket* vehicle summary.

Parameter	Specification
Dry Mass (unballasted / ballasted)	37.2 / 38.2 lbm
Wet Mass (unballasted / ballasted)	41.1 / 42.1 lbm
Diameter	6.17 in
Length	97 in
Center of Gravity	53.3 in (from nose tip)
Center of Pressure	66.7 in (from nose tip)
SSM (On Pad)	2.18 cal
Motor	Cesaroni L1720
Main Parachute	96 in Fruity Chutes Iris Ultra
Drogue Parachute	18 in Fruity Chutes Classic Elliptical

1.3 Payload Summary

The Extraterrestrial Lander for Data Collection and RF Transmission is a parachute-arrested lander that acts as the nose cone of the launch vehicle and jettisons at 300 ft AGL during main parachute descent. Jettison is enabled by electromechanical latches that release the payload, allowing gravity to pull it from the foresection of the launch vehicle. Once jettisoned, the payload deploys a parachute and landing legs that double as the ground plane for the antenna system. The payload lands horizontally (nose cone parallel to the ground) upon which the parachute is detached. Fifteen seconds later, the antenna is raised to a height of 24 in above the ground allowing the payload to begin RF transmission of flight telemetry. The payload is constructed out of aluminum and 3D printed bulk plates attached to 2 aluminum rods to transfer load throughout the structure. The airframe and nose cone are made of fiberglass to reduce RF interference and are fixed to the skeleton via two bulk plates. The fore bulk plate contains 3 radial bolts while the aft bulk plate slots into a retention ring epoxied to the airframe itself. The legs and antenna are deployed through slots cut into the airframe. The nose cone is attached to the airframe directly via four radial bolts.

2 Assessment of Flight



Figure 1: Fully assembled and painted vehicle on competition day.

2.1 Flight Mission Criticality

Table 3 outlines the mission success criteria and relevant requirements pertaining to the flight itself.

Table 3: Flight mission success criteria, relevant requirements, and statuses.

Success Criterion	Relevant Requirement	Status
Achieve L.R.E. velocity of >52 fps	NASA 2.17	Met
Obtain thrust to weight ratio of >5.0:1.0	NASA 2.15	Met
Achieve apogee >3500 ft	NASA 2.1	Met
Deploy drogue parachute <2 seconds following apogee	NASA 3.1.2	Met
Deploy main parachute >500 ft	NASA 3.1.1	Met
Drift <2500 ft	NASA 3.11	Met
Achieve L.K.E. <75 ft-lbf for all vehicle sections	NASA 3.3	Met
Descend from apogee in <90 seconds	NASA 3.12	See Section 2.3.

2.2 Flight Sequence

Figure 2 captures key flight events, timestamps, and values from the competition flight.

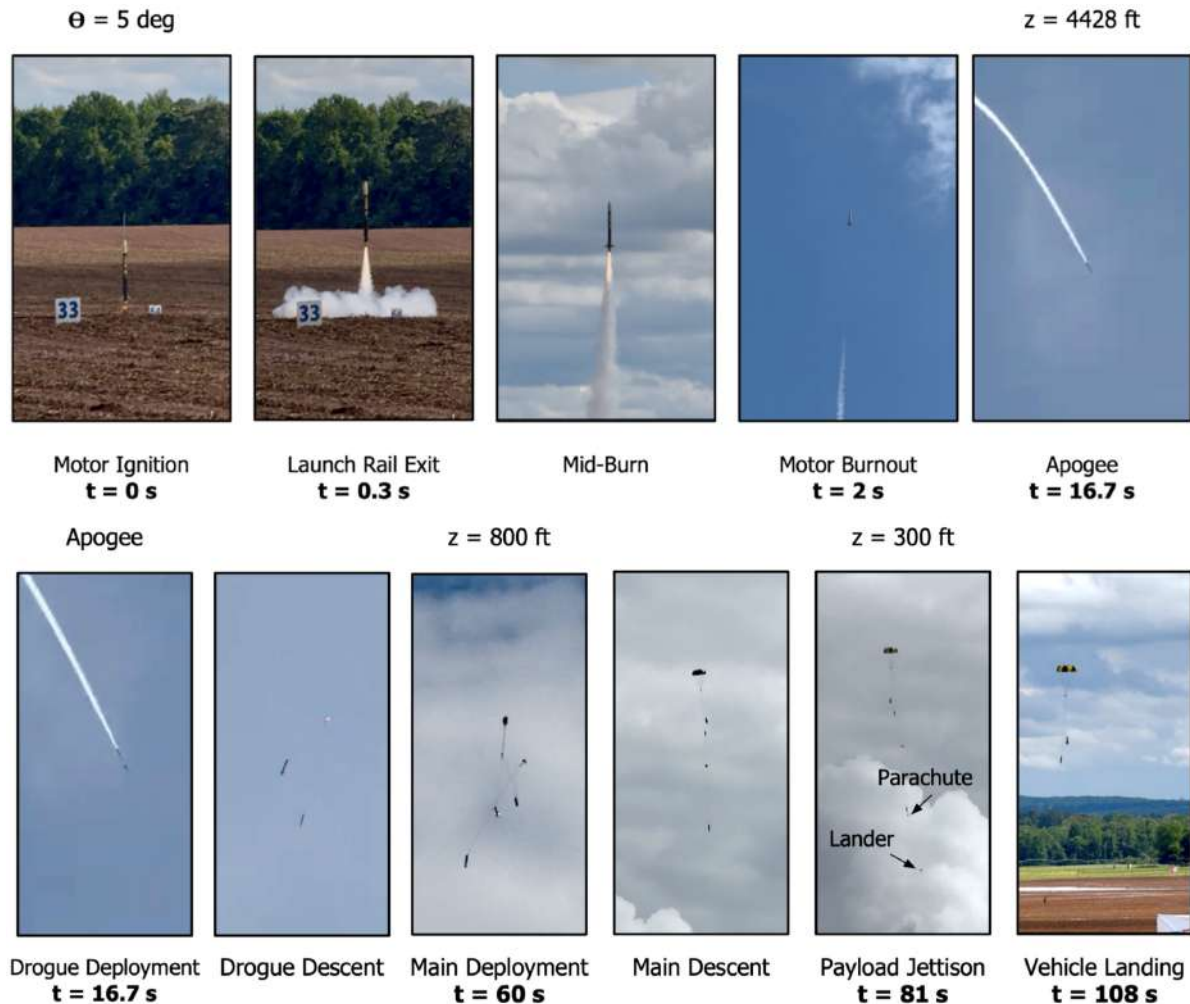


Figure 2: Key flight events, values, and timestamps.

2.3 Flight Performance

The VADL 2025 USLI Flight achieved nominal performance in all aspects of vehicle ascent, which are summarized below in Table 4. The 6.6 mph sustained wind speed presented on launch day was utilized in the following pre-flight model simulations conducted at 6:00 AM, May 4, 2025 CDT.

Table 4: Vehicle ascent performance relative to pre-flight model predictions.

	Apogee (ft)	Predicted Rate (fps) & Relative Error	Launch Rail Exit Velocity (fps)	Predicted LRE Velocity (fps) & Relative Error	Motor Delivered Impulse (N-s)	Predicted Impulse (N-s) & Relative Error
VADL USLI	4428 (Main) 4429 (Backup)	4448 -0.4%	78.4	78.8 -0.5%	3572	3566 0.2%

The pre-flight model follows methodology extensively detailed in the Mission Performance Predictions (MPP) Section in VADL's FRR Milestone Report using environmental measurements gathered on launch grounds, which are detailed in Table 5.

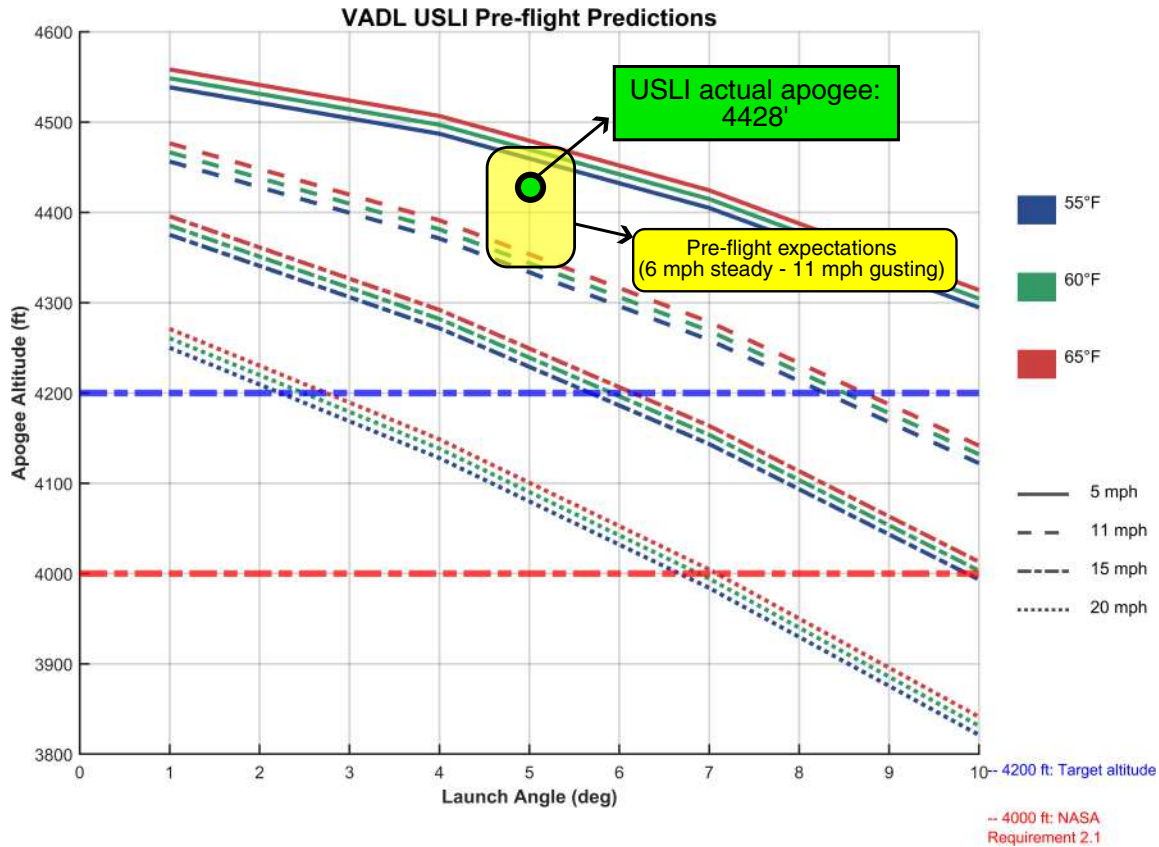


Figure 3: Pre-flight apogee reference chart at the point of Huntsville LRR.

Using Table 5, the pre-flight model predictions outlined in Table 4 align remarkably well with the measured ascent metrics in apogee achieved, launch rail exit velocity, and motor impulse delivery. This strongly suggests that the VADL competition vehicle was well built up-to-spec and performed without any subnominal events.

One day prior to the USLI Competition, VADL had utilized measured massing of the competition vehicle to produce a reference sheet that relates launch-day rail angle and wind speeds with predicted apogee. The achieved apogees from raw altimeter data (Figure 8) show high alignment within the predicted regimes of apogee outcomes (Figure 3).

Table 5: In-field measurements utilized for the pre-flight model predictions.

In-field Measurements	Value
Wet Mass (lbm)	42.1
Wet CG (in)	53.3
CP (in)	66.7
SSM on Pad	2.18
Ground Wind Speed (mph)	6.6 (sustained) 11 (gust)
Launch Rail Angle (deg)	5
Launch Site Elevation (ft)	800
Launch Site Temp. (F)	60
Ballast (lbm)	1.0
Effective Launch Rail Travel (in)	144

Consistent with the methodology described and extensively justified in the FRR Report, LRE velocity was determined by integrating payload IMU acceleration, demonstrating the nominal performance of the L1720 motor (Figure 4). As detailed in the FRR Report MPP, a literature-informed friction factor of 3.5% has been applied to all pre-flight models. While the VADL VDFs measured a lower friction factor of approximately 1.5%, this discrepancy is attributed to (1) non-existent cross winds at the VDF site and (2) the VDF rails being well-lubricated with a liberal amount of WD-40 water-based penetrating oil, which significantly reduced rail friction but was absent on the competition rail. The USLI apogee and motor thrust measurements (Figure 5b) further validate the application of the 3.5% friction factor recommended by current literature.

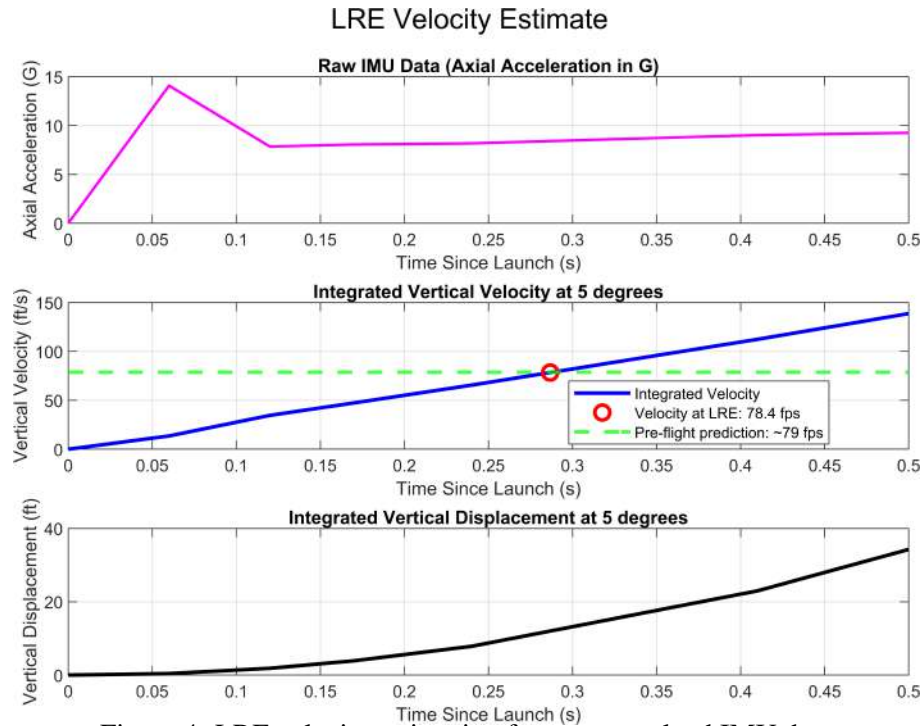
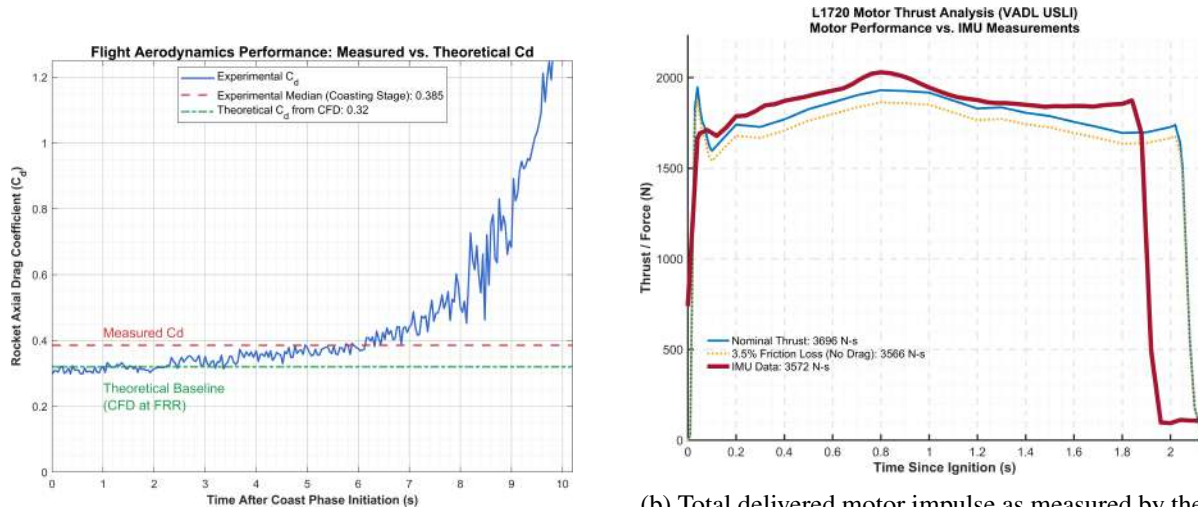


Figure 4: LRE velocity estimation from raw payload IMU data.

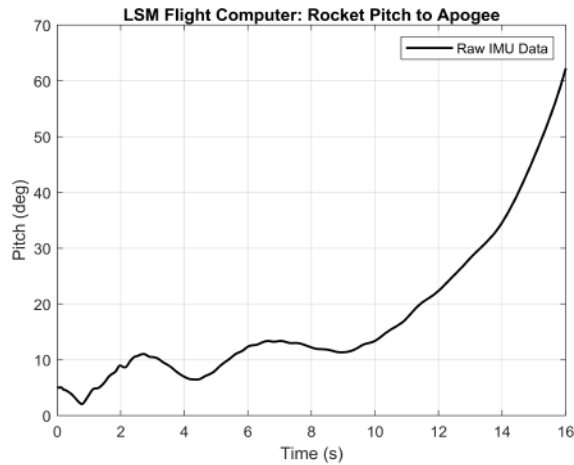
Using the coasting phase drag coefficient estimation methodology extensively detailed and justified in VADL's FRR Milestone Report, the experimental drag coefficient closely aligns with pre-flight CFD simulations (Figure 5a), which was extensively validated in the VADL VDFs and PDF.



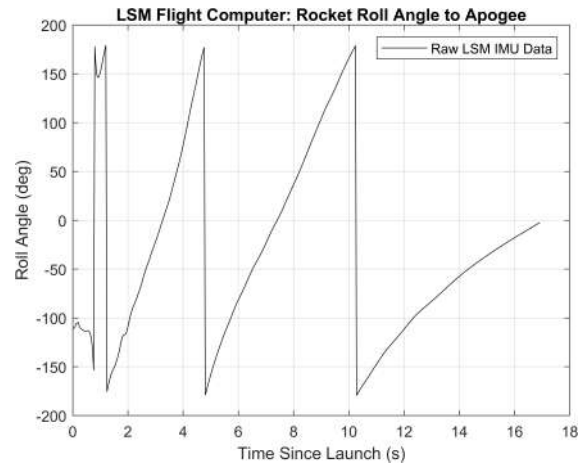
(a) Vehicle aerodynamic performance.

(b) Total delivered motor impulse as measured by the LSM Flight Computer IMU.

IMU gyroscopic data computed rocket pitch and roll shown in Figures 6a and 6b, respectively. The relatively low number of oscillations in rocket pitch suggest minimal wind negotiation in the first few seconds of flight, consistent with the low sustained wind conditions on launch day. Neither roll nor pitch data suggest issues in vehicle aerodynamic stability, structural rigidity, asymmetrical thrust, or fin fluttering concerns. Coupled with a nominal delivery of motor impulse, nominal aerodynamic performance and an apogee outcome that was well within pre-flight predictions, IMU gyroscopic data supports the notion that the VADL USLI Flight was successful in all aspects of vehicle ascent metrics. Velocity versus time data is shown in Figure 7.



(a) Vehicle pitch to apogee.



(b) Vehicle roll to apogee.

Figure 6: LSM flight computer angular data.

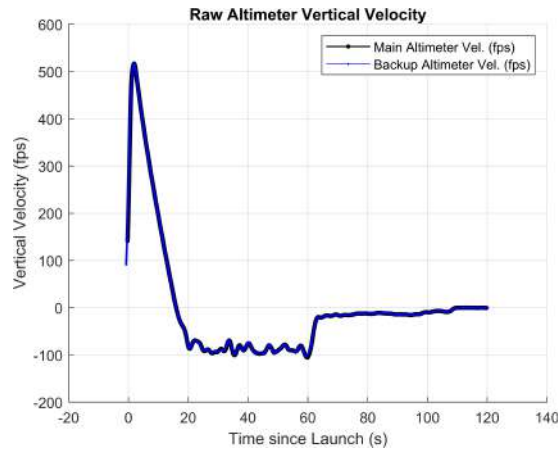


Figure 7: Avionics altimeter raw velocity versus time data.

2.4 Recovery Performance

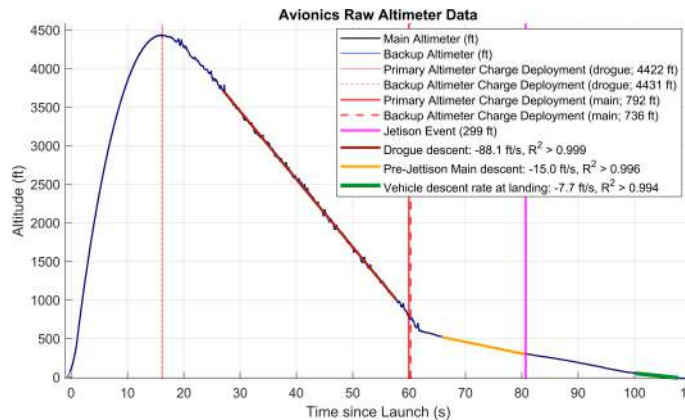


Figure 8: Avionics altimeter raw altitude versus time data.

Both the jettisoned payload and the rocket achieved nominal recovery in mission-critical charge deployments as well as landing kinematics and kinetics, which are summarized in Figures 8 and 10 and Tables 7 and 8. Landing kinetic energies of independent vehicle sections are

Table 6: Landing kinetic energies at landing.

VADL USLI	Sectional KEs at Landing (lb-ft)			
	Fore	Mid	Aft	Payload
VADL USLI	6.8	11.6	9.8	16.5

a most critical indicator of recovery performance. Impressively, the highest sectional landing kinetic energy recorded was 16.5 lb-ft (payload), significantly below both the 75 lb-ft threshold specified by NASA Requirement 3.3 and the stricter 65 lb-ft bonus criteria (Table 6).

Table 7: Recovery performance of payload and rocket as derived from avionics raw altimeter data and payload IMU raw altimeter data.

	Drogue Descent Rate (fps)	Vehicle Descent Rate at Landing * (fps)	Payload Descent Rate at Landing * (fps)	Vehicle Descent Time (s)	Payload Descent Time (s)
VADL USLI	88.1	7.7	11.9	92.3	82.4
* linear regression of altimeter data for the last 50 ft of descent ($R^2 > 0.98$)					

Notably, the vehicle's descent time of 92.3 s highlights the nondeterministic impact of environmental wind conditions during launch, which fell outside the regime of pre-flight predictions. Figure 9 illustrates an unexpected upward wind gust at 60 ft AGL that reduced the rocket's final landing velocity to an unusually low 7.7 fps. Consequently, the descent time extended from the expected no-gust descent duration of 88 s to 92.3 s, slightly exceeding the 90 s threshold stipulated in NASA Requirement 3.12. Excluding this statistical and environmental anomaly, VADL maintains confidence that the recovery system was fully designed to comply with this requirement.

Table 8: Mission-critical deployment altitudes as reported by the embedded softwares in the primary avionics altimeter and the LSM flight computer.

	Drogue Deployment Altitude (ft)	Main Deployment Altitude (ft)	Payload Jettison Altitude (ft)
VADL USLI	4422	792	299

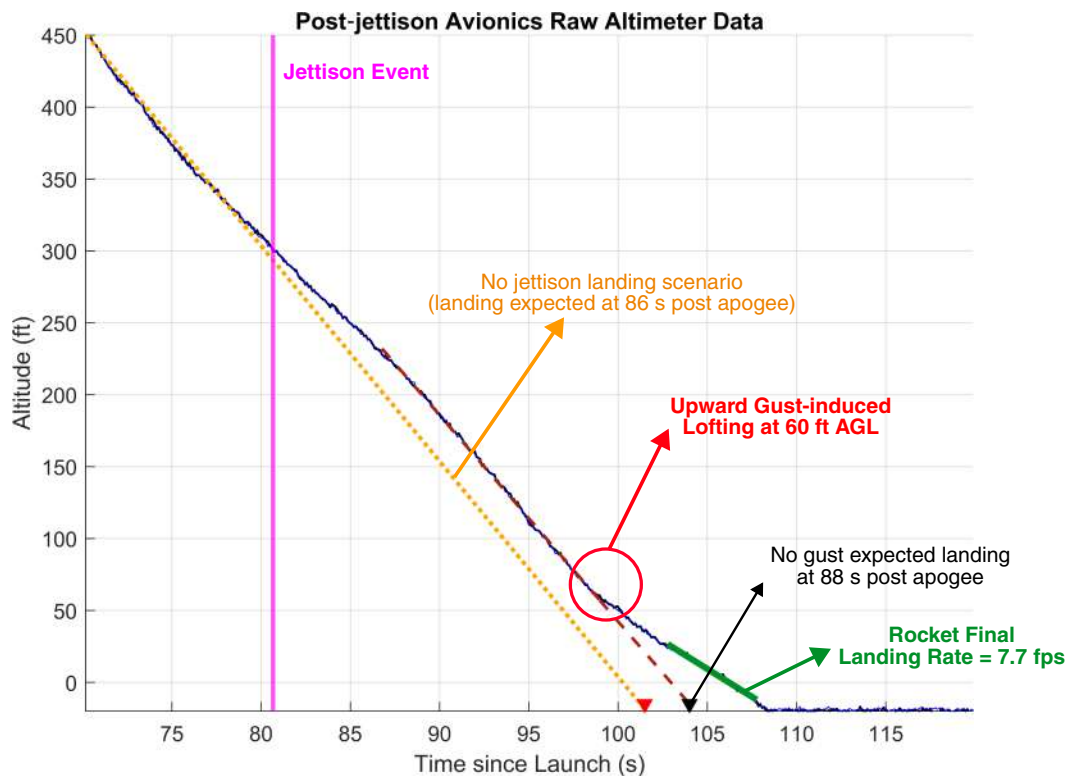


Figure 9: Post-jettison avionics raw altimeter data, showing effects of upward gust-induced lofting at 60 ft AGL.

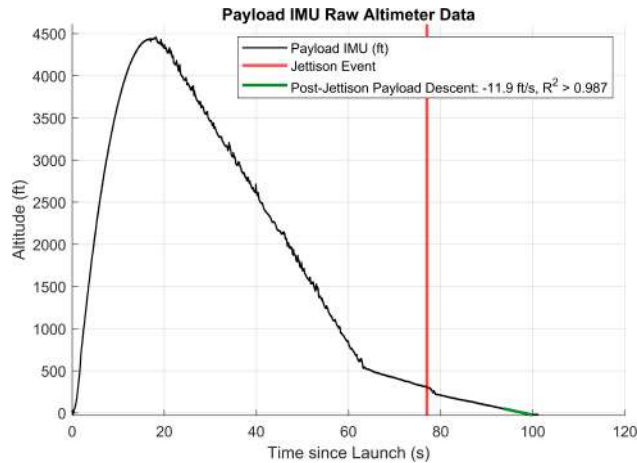


Figure 10: Raw payload IMU altimeter data for payload landing analysis.

landed 3107 feet away from the NASA receiver. This distance falls within the testing parameters of successful transmission, which has been proven at distances up to 1 mile.

Shown in Figure 10 is the payload IMU raw altimeter data, further validating the flight analysis laid out in the preceding sections. As seen, the payload descended at a rate of 11.9 fps, well within pre-flight simulations.

2.5 GPS Data

The vehicle and payload recovery GPS modules were positioned in the LSM bay and the payload, respectively, and stored GPS data at a rate of 1 Hz. Figure 11 highlights the landing site, launch pad, Vanderbilt receiver, and NASA receiver, visualized using Google Earth. Also shown are the distances between these points and the payload landing site. Notably, there was 1095 feet of drift between the launch pad and the landing site, and the payload

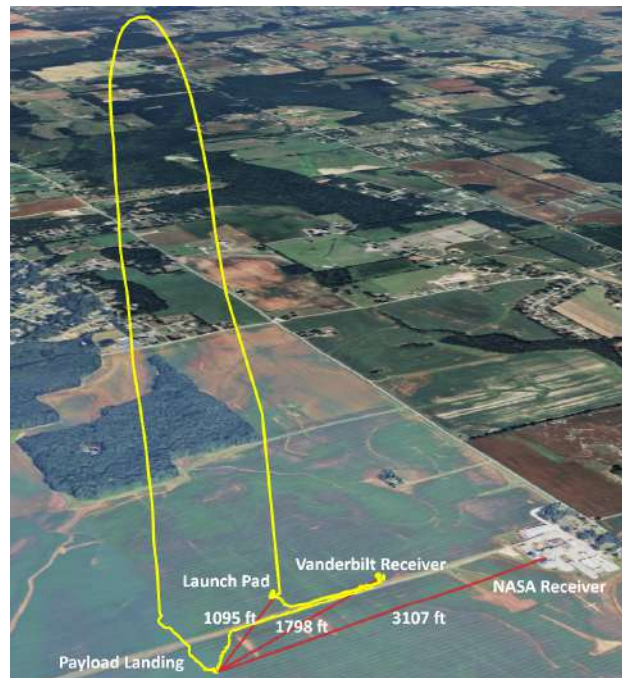


Figure 11: GPS flightpath and distances to receivers.

3 Assessment of Vehicle

3.1 Vehicle Mission Criticality

Table 9 outlines the mission success criteria and relevant requirements pertaining to the launch vehicle. All criteria were successfully met.

Table 9: Vehicle mission success criteria, relevant requirements, and statuses.

Success Criterion	Relevant Requirement	Status
Prepare the vehicle in <2 hours	NASA 2.5	Met
Possess static stability margin of >2.0 cal at rail exit	NASA 2.14	Met
Retain payload until jettison	NASA 4.3.4	Met
Incur no damage to any vehicle sections	NASA 2.3	Met

3.2 Vehicle Design Overview

The vehicle shown in Figure 12 was flown at the USLI competition. The specific sectional lengths, separation points, and recovery locations are labeled. The overall vehicle length was 97 inches, mass was approximately 42.1 lbm, and powered by a Cesaroni L1720 motor. The recovery system included a 96-inch main parachute, an 18-inch drogue parachute, and a payload parachute. The main and drogue recovery systems were deployed using black powder while the lander recovery was enabled through electromagnetic latches. The landed masses of the payload, fore, mid, and aft sections were 7.5 lbm, 7.4 lbm, 12.6 lbm, and 10.7 lbm, respectively.

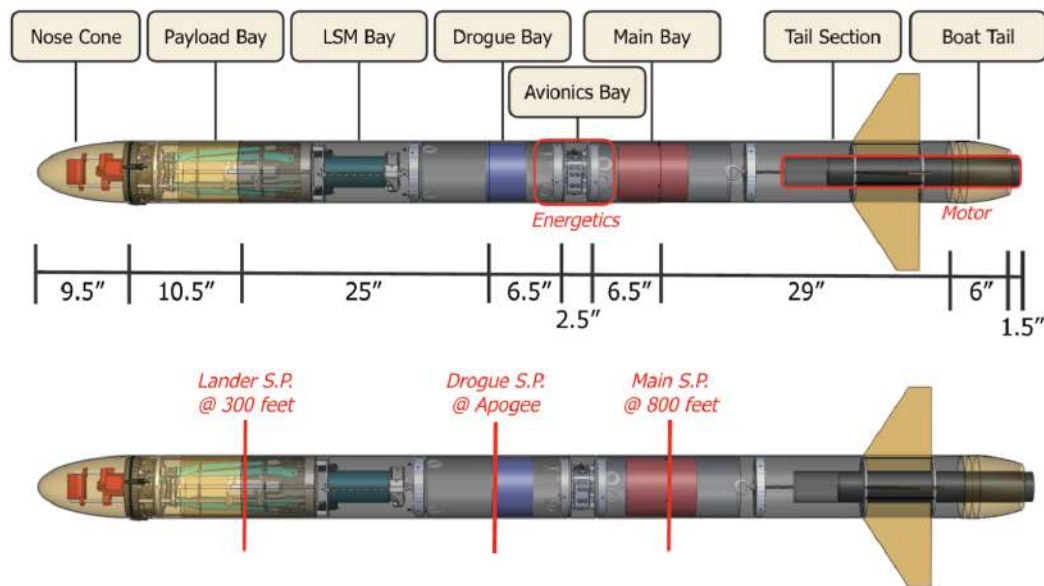


Figure 12: Competition vehicle breakdown.

3.3 Vehicle Preparation

Figure 13 outlines the preparation and assembly timeline for the launch vehicle on competition day. The vehicle was prepared in 1.5 hours, from 12:30 to 2:00 PM, thus satisfying NASA Requirement 2.5.

Time	8:00-8:45 AM	8:45-9:45 AM	9:45 AM - 12:30 PM	12:30-1:00 PM	1:00-1:30 PM
Event	Arrived at Bragg Farms, unloaded supplies, setup tent, tables, and chairs	Performed final payload testing, inspected all vehicle and payload hardware for damage from transportation	Holding period due to weather delays	Assembled motor	Assembled recovery hardware and avionics
Time	1:30-2:00 PM	2:15-2:30 PM	2:30-2:50 PM	3:30 PM	3:50-4:15 PM
Event	Completed full assembly of vehicle and payload	Pre-flight RSO inspection	Setup vehicle on launch rail, armed altimeters, and installed igniter	Launch	Recovered vehicle and payload sections

Figure 13: Competition vehicle breakdown.

3.4 Vehicle Reflight Certification

The landed configurations of all vehicle sections are shown in Figure 14 below. As seen, no damage was incurred to any section.

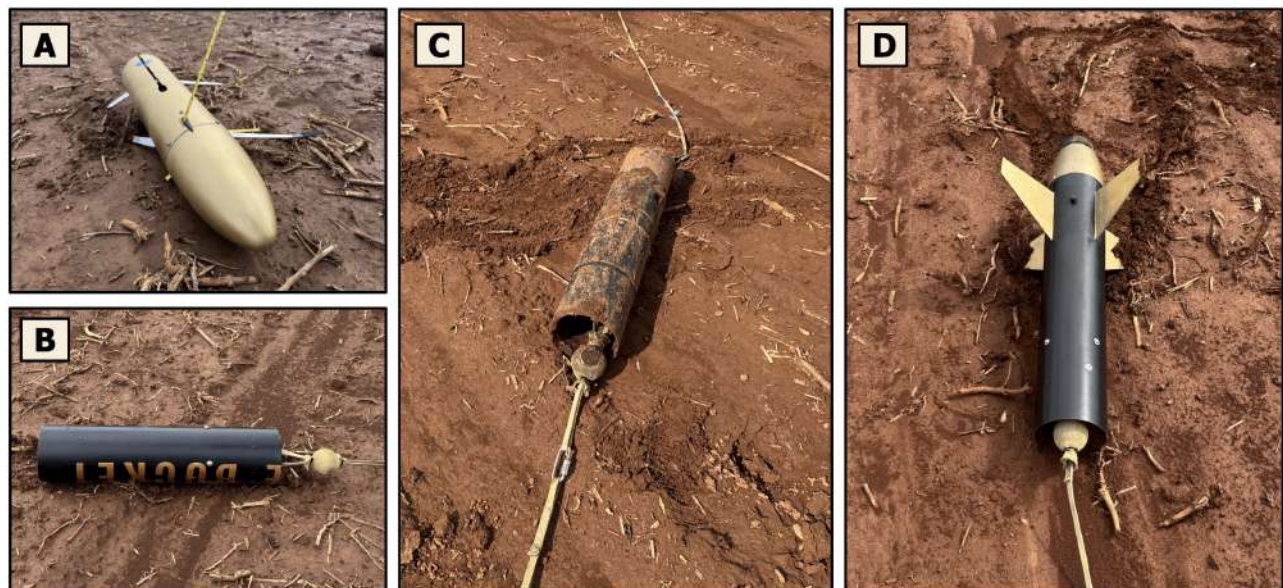


Figure 14: Landed configurations of (A) payload, (B) LSM bay, (C) recovery section, and (D) tail section.

To further validate the ability to reflly the vehicle, a structural load analysis was performed on key vehicle components. Payload IMU data was used to obtain the acceleration forces experienced at each parachute deployment event. Then, the factor of safety was calculated for the lowest-rated component. The results are compiled in Table 10 below, showing that no component experienced a FoS less than 5.

Table 10: Lowest factor of safety for each parachute deployment event.

Event	Acc (ft/s ²)	G's	Max Force (lbf)	FoS	Component / Shock Rating (lbf)
Drogue	472	14.7	342	8.8	Swivel Link / 3000
Main	689	21.4	589	5.1	Swivel Link / 3000
Payload	329	10.2	76.6	9.1	Shock Cord / 700

A summary of the post-flight inspection results is shown in Table 11. A comprehensive breakdown of the reverification information is provided in Appendix A.

Table 11: Inspection and reflight verification results for key component groups

Component Group	Inspection Result	Reflight Verification Result
Airframes	No Damage	Verified
Recovery	No Damage	Verified
Couplers	No Damage	Verified

3.5 Vehicle Lessons Learned

The USLI challenge instilled precision and good engineering practices within the team at every milestone of the project cycle. The team gained significant experience with design, fabrication, and testing the vehicle and mission-critical components. Some of the lessons learned are listed below:

Table 12: Key project management lessons learned and how they were addressed.

Lesson Learned	Mitigation Method
Standardization of tools and fasteners is essential to maximizing efficiency.	The fasteners, bolts, washers, etc., were standardized to ensure that the vehicle construction efficiency was maximized. It also allowed for faster vehicle assembly/disassembly on launch day, enabling the team to assemble the launch vehicle in under 30 minutes.
Following protocols for fabrication is of utmost importance to ensure precise manufacturing.	The team emphasized safety measures, design restrictions imposed by NASA, and the fabrication protocols developed by the team while constructing the launch vehicle through well-defined responsibilities and task lists. This reduced the possibilities of human error while also making it easy to track back to any steps missed, which could have otherwise resulted in imprecise manufacturing.
Rigorous testing must be implemented to ensure the structural integrity of the launch vehicle.	The team implemented proper testing of not only mechanisms but also materials, airframe, bulk plates, housings, etc., all critical to the safety of the vehicle. This testing provided great information on material strength to validate analytical results provided by FEA and hand calculations.
For precise coupler fitting, the intentional placement of temporary surface-masking tape must be applied to avoid unnecessary epoxy, paint, etc.	The team learned early on the disadvantages of unnecessary epoxy, paint, etc. To mitigate this issue for subsequent couplers, the team added a step to wrap the non-shoulder portions of the couplers with painter's tape to ensure precise fitting of the coupler into the airframe.

4 Assessment of Payload

4.1 Payload Mission Criticality

Table 16 outlines the mission success criteria and relevant requirements pertaining to the flight payload. As shown, all criteria were successfully met.

Table 13: Payload mission success criteria, relevant requirements, and statuses.

Success Criterion	Relevant Requirement(s)	Status
Retain STEMnauts through flight	NASA 4.1; VADL 4.1.1	Met
Jettison at 300 ft	VADL 4.2.1-4	Met
Passively deploy legs	VADL 4.3.1-3	Met
Land upright	VADL 4.1.7	Met
Experience DRI of <17.7	VADL 4.1.6	Met
Detect landing	VADL 4.7.2	Met
Detach parachute	VADL 4.4.1-2	Met
Extend antennas	VADL 4.5.1-5	Met
Be reflyable following landing	VADL 4.1.8	Met

4.2 Design Overview

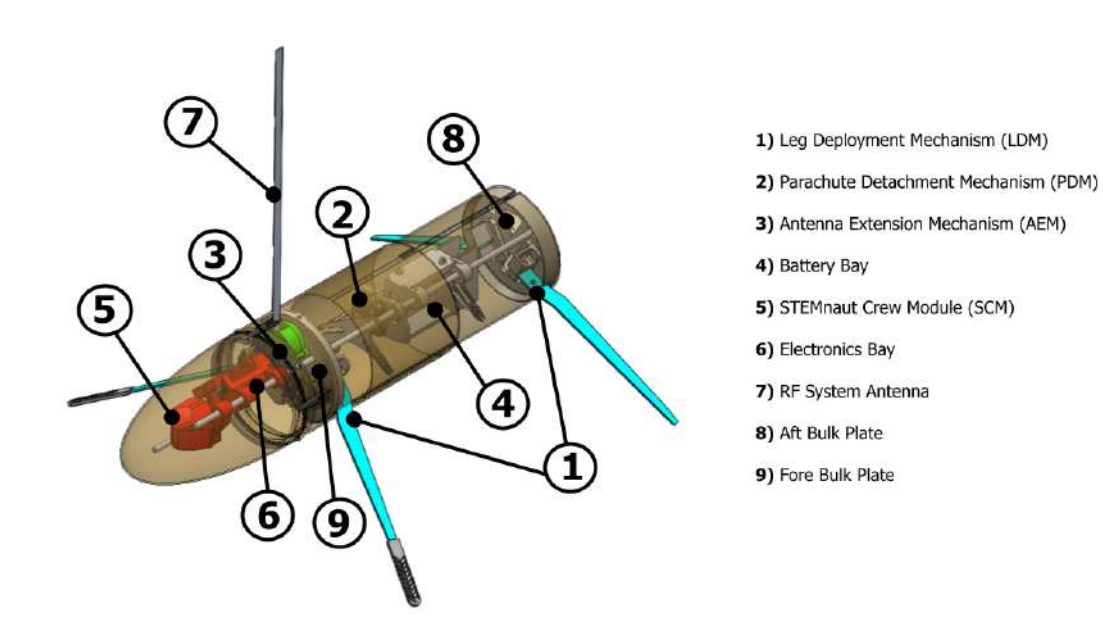


Figure 15: Payload overview with key mechanisms highlighted.

The internal structure of the payload can be seen in Figure 15 above. The landing legs are attached to the payload on the fore and aft bulk plates and are deployed by torsional springs upon jettison. The PDM enables the release of the parachute upon landing, which is facilitated by an electromagnetic latch. The AEMs deploy 15 seconds after landing, and RF transmission begins after this extension. The electronics

and SCM are housed in the nose cone for easy access, and the battery bay is located in the aft end of the payload to provide the desired mass distribution. The STEMnauts consist of four lego minifigures.

4.3 Payload Retention and Detachment

The payload is retained through the use of two rotary latches in the LSM. The latches are Southco R4-EM-05D00-150 electronic rotary latches and are rated for up to 1061 lbf of tensile load before failure. The EM latches are attached to a 0.15in thick aluminum bulk plate located at the bottom of the LSM assembly via two aluminum blocks. All fasteners connecting the EM latches, blocks, and bulk plate are steel 8-32 bolts.



Figure 16: LSM latch to bulk plate retention.

The payload interfaces with the EM latches through two steel hooks that are fastened to the aft payload bulk plate through steel 8-32 bolts. The aft bulk plate is fabricated from 6061 aluminum and is 0.25in thick. This aft bulk plate is retained to the payload skeleton through two 5/16-24 nuts.

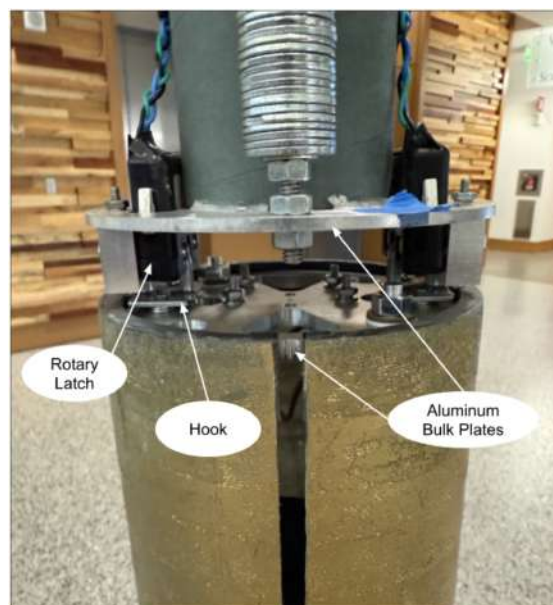


Figure 17: Payload retention system.

The payload is detached by opening the EM latches upon receiving RSO approval through a TBS Crossfire transmitter that sends an RF signal to the LSM computer over the 915 MHz band. This approval is then checked against the onboard altimeter to ensure payload jettison occurs at 300 ft AGL.

4.4 Mechanism Performance

All key mechanisms performed nominally with the exception of partial antenna extension for the AEM. The LSM successfully retained the payload throughout launch and recovery events before releasing the payload at 300 ft AGL autonomously after receiving RSO approval. The PDM withstood the rigors of parachute opening and landing without premature release of the parachute. Upon landing the PDM successfully detached the parachute to prevent the lander from toppling over. The LDM successfully deployed the legs following payload jettison and allowed the lander to maintain an upright orientation upon landing. The AEMs successfully extended all three antennas following landing. It was noted upon inspection that the top antenna did not extend completely to 18 in, and only extended to 12 in.

The team has verified that robust RF transmission is possible even with non-extended antennas and conducted a post-flight certification test outlined in Section 4.5 to test the as landed configuration. Thus, all mechanisms performed successfully to complete all aspects of the payload mission from jettison, to landing and parachute detachment, to RF transmission.

4.5 Payload Reflight Certification

Post-flight inspection of the payload confirmed that all mechanical subsystems remained fully intact and functional, with no visible damage or missing components (Section 4.4). On the electronics side, the nose cone was removed after landing to allow inspection of the wiring and embedded systems. All components remained securely housed, the flight computer was still running with the cooling fan active, and no disconnected or floating wires were observed, as shown in Fig. 18.

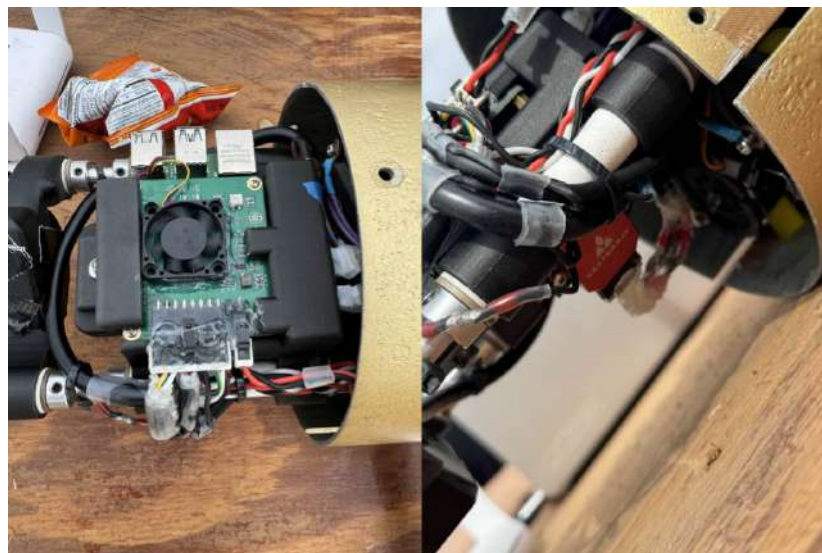


Figure 18: Payload electronics immediately after landing. All systems remained intact; the cooling fan was active, and no wires were disconnected or dislodged.

After returning home, the team conducted a flight test at Centennial Park to further verify functional reliability. The test configured the system to immediately detect landing and trigger APRS transmission. As

expected, the system performed robustly, and data was correctly transmitted, confirming that it remains flight-ready.



Figure 19: Centennial Part Test. The experiment is set up to mimic the competition day situation.

It is important to note that for the Centennial Park test, the payload was set up in the exact configuration as during the competition day, after it landed. The receiver station, using a Diamond X50 antenna and RTL-SDR, was set at approximately the same distance between the ground station and payload during competition day.

A summary of the post-flight inspection results is shown in Table 14. A comprehensive breakdown of the reverification information is provided in Appendix A.

Table 14: Inspection and reflight verification for key payload mechanisms and components.

Component Group	Inspection Result	Reflight Verification Result
Airframe	No Damage	Verified
Nosecone	No Damage	Verified
Bulk plate and Skeleton	No Damage	Verified
Electronics and RF System	No Damage	Verified
AEM	No Damage	Verified
PDM	No Damage	Verified
LDM	Replaceable Shoes Deformed, Legs Undeformed	Verified

4.6 Payload Lessons Learned

The payload is a joint effort between the Electrical and Mechanical subteams. It integrates sensing, control, and deployment mechanisms into a unified system. Clear communication between the electrical and mechanical engineers was prioritized throughout development. The design emphasizes simplicity, with passive solutions implemented wherever feasible to increase robustness and reduce failure points. For example, both the antenna and leg deployment systems use passive mechanisms, which proved to be highly reliable throughout the project. The team observed that simpler systems tend to perform more consistently under real-world conditions. Additionally, extensive testing ranging from RF transmission range tests to payload functionality drop tests was conducted to ensure functionality post-flight and post-landing. This high volume of testing enabled the team to drive design iterations informed by actual results, giving the team a taste of what the Aerospace industry is like. Along the way, many lessons were learned by working through challenges that arose. Several of these lessons and their mitigation methods are captured in Table 15 below.

Table 15: Key payload lessons learned and how they were addressed.

Lesson Learned	Mitigation Method
Interdisciplinary integration takes more time than anticipated	Design reviews included representatives from all subteams to identify cross-system dependencies early in the cycle.
Overcomplicated systems increase debugging time and risk of failure	A design philosophy of “simplicity first” was adopted. This happened after the “Vertical Lander” failed.
Limited documentation slowed the onboarding of new team members	A documentation sprint was added after CDR to capture system diagrams, wiring schematics, code bases, and configuration procedures.
Insufficient RF shielding caused the I ² C signal to be unreliable	Aluminum shielding and grounding paths were added to the PCB and wiring systems.
Testing without environmental variation limits failure detection	All subsystems were retested under temperature and vibration profiles similar to launch and landing conditions.
Payload fit-checks were delayed due to incomplete CAD integration	A full CAD integration checkpoint was scheduled three weeks before FRR to allow time for adjustments.

5 Assessment of Experiment

5.1 Experiment Mission Criticality

Table 16 outlines the mission success criteria and relevant requirements pertaining to the payload experiment.

Table 16: Payload mission success criteria, relevant requirements, and statuses.

Success Criterion	Relevant Requirement(s)	Status
Collect flight telemetry and landing site data	NASA 4.2.1; VADL 4.7.1-9	Met
Transmit radio frequency data on the 2-M band	NASA 4.2.1, 4.2.3, 4.2.4, 4.2.6; VADL 4.1.3, 4.6.1-3	See Section 5.3.

5.2 Software Performance and Data Collection

```
IMU initialized, starting data monitoring
DEBUG: Above launch threshold (547m). Starting launch timer...
DEBUG: Launch timer reset (Altitude dropped to 546.49 m)
DEBUG: Above launch threshold (547m). Starting launch timer...
DEBUG: Launch timer reset (Altitude dropped to 542.45 m)
DEBUG: Above launch threshold (547m). Starting launch timer...
DEBUG: Launch detected at t=4398.39s (Altitude: 831.62 m)
DEBUG: Start freeze period
DEBUG: End freeze period
DEBUG: Low altitude detected
DEBUG: New minimum altitude recorded: 265.73 m at t=4485.41s
DEBUG: Altitude violation (Altitude: 267.76 m) - resetting min altitude.
...
DEBUG: New minimum altitude recorded: 263.77 m at t=4485.85s
...
DEBUG: New minimum altitude recorded: 225.41 m at t=4493.43s
RF Data Collected.
DEBUG: Landed state detected after stable low altitude of 5.0 s.
[survivability] a=-76.60 v=4.40 DRI=4.51
Data logging process started.
Landing detected. Starting post-landing timeout.
Timeout reached. Stopping logging process.
Data logging completed. Logs combined into data_log_combined.txt
DEBUG: Detach Parachute

Stopping processes...
Closing I2C bus...
GPIO pin released and handle closed.
Processes stopped. GPIO cleanup done.
Closing serial connection...
```

Figure 20: Software output log showing flight states.

The software log (Fig. 20) was examined post-flight. Launch was detected at $t = 4398$ s (system time) at an altitude of 2729.66 ft (831.62 m), followed by a successful exit from the freeze period. Low-altitude tracking is activated with a minimum altitude threshold of 739.53 ft (225.41 m). Landing was confirmed after maintaining stability for 5 seconds, after which the APRS transmission was activated 15 seconds later. The data transmitted is shown in Table 17. The data logger operated correctly, and output was merged into `data_log_combined.txt`. A parachute detach command was issued, and the system completed a clean shutdown with all I²C, GPIO, and serial interfaces properly closed. No errors were logged.

The data for transmission on APRS are shown in Table (17). Important to note is the DRI value of 4.51, well below the threshold of 17.7 for lowest injury risk. This result confirms a safe flight and landing for the STEMnauts.

Table 17: Transmitted Data. STEMnaut Bay Temperature and Battery State of Charge updates in real-time, and may differ from landing data.

Symbol	Meaning	Data To Transmit
T	STEMnaut Bay Temperature	93.21 °F (On Landing)
A	Apogee Altitude	4452.53 ft
B	Battery State of Charge	84.51% (On Landing)
S	Survivability Index	4.51 DRI
Qw, Qx, Qy, Qz	Orientation	-0.56, 0.49, -0.53, -0.41
LandingT	Landing Time	15:32:30
LG	Landing Acceleration	-88.29 m/s ²
MaxV	Maximum Velocity	576.67 ft/s
LV	Landing Impact Velocity	14.44 ft/s

5.3 Protocol and Interference Risks

All 8 data points were successfully transmitted during our flight, verified by software (Section 5.2), and confirmed with a handheld receiver in the field. On the payload side, post-landing inspection showed that the antenna was deployed correctly and hardware and wires were secured and intact (Section 4.4). Additionally, the re-flyable test replicating the competition day payload configuration and transmission range was also performed after the competition to show that all systems are operational (Section 4.5).

However, due to unexpected cloudy weather, a short launch window, and scheduling constraints, five teams were launched within a 10-minute window to ensure all remaining teams had the opportunity to fly before conditions changed. This compressed launch cadence resulted in overlapping rocket flights. During our specific 10-minute landing window, five teams, including Vanderbilt, were likely transmitting APRS packets simultaneously on the same 2M frequency (Figure 21).

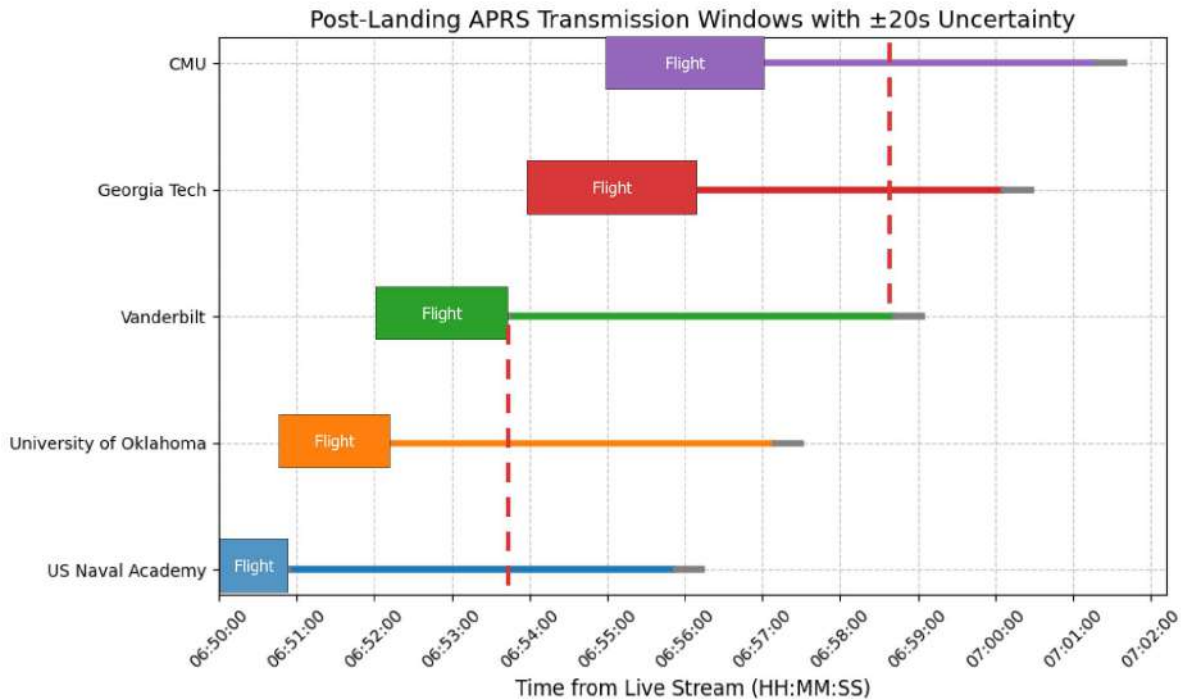


Figure 21: Timing diagram showing post-landing APRS transmissions within a 10-minute window, with Vanderbilt positioned in the middle. The diagram includes estimated flight and transmission times, derived from the live-stream recordings of NASA, for the U.S. Naval Academy, the University of Oklahoma, Vanderbilt University, Georgia Institute of Technology, and Carnegie Mellon University. The red dashed lines indicate Vanderbilt's 5-minute transmission window, which overlaps with transmissions from the other four teams.

This raises concerns regarding RF congestion. APRS operates on an ALOHA-like protocol, which lacks collision avoidance (Kenneth W. Finnegan, Examining Ambiguities in the Automatic Packet Reporting System, California Polytechnic State University [1]). The Finnegan thesis specifies that collisions result in all packets being lost, with no feedback (unlike CSMA/CD or other more robust schemes). While VADL's software and hardware confirmed packet generation and transmission, overlapping broadcasts could lead to packet loss at the receiver end.

VADL respectfully request that NASA review whether concurrent APRS transmissions from multiple teams can be reliably received and decoded, especially in high-traffic landing windows. Despite correct system operation on Vanderbilt's end, protocol-level limitations may compromise successful packet reception under congested conditions.

5.4 STEMCRaFT Descent and Landing Analysis

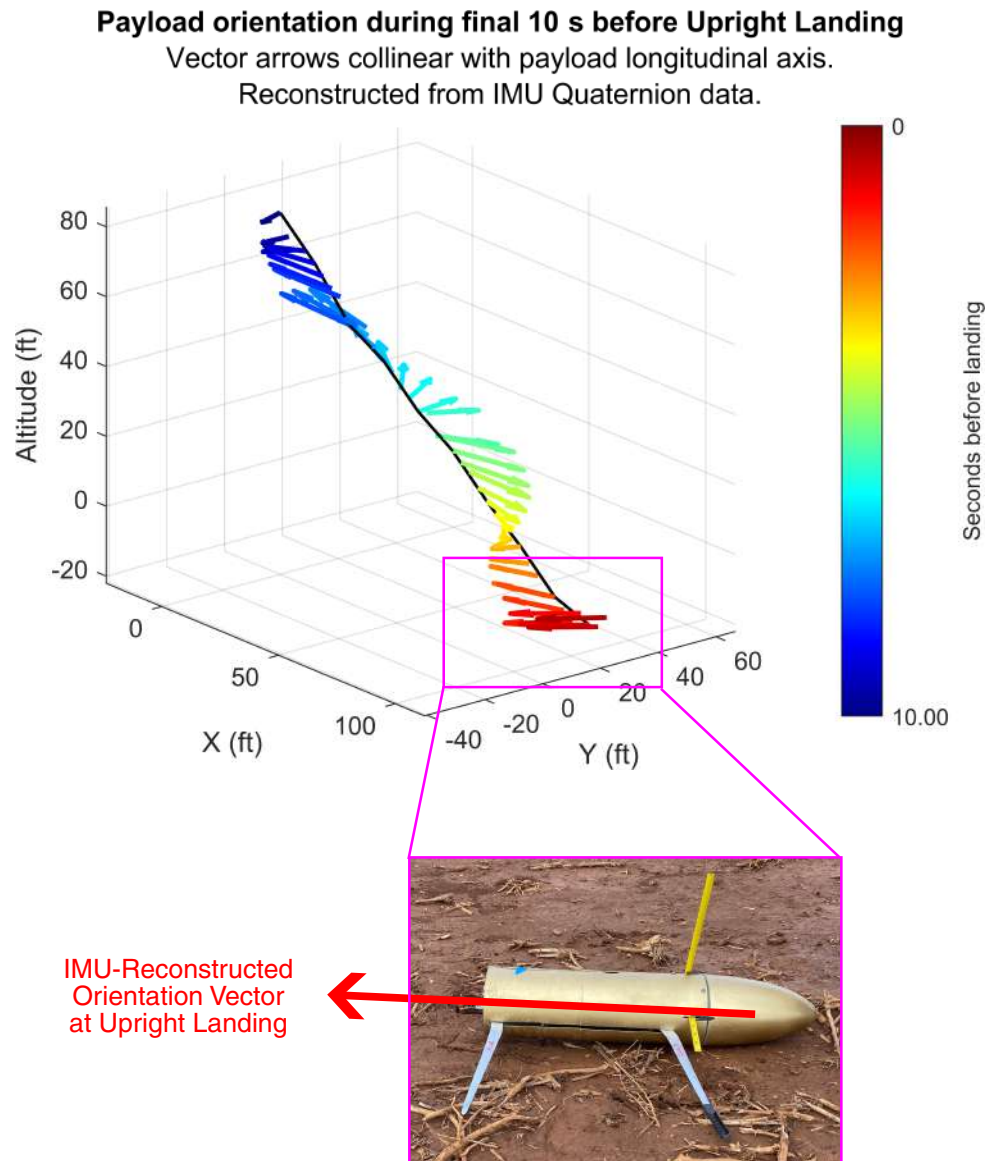


Figure 22: Stable STEMCRaFT descent and upright landing as shown from IMU-reconstructed time evolution of STEMCRaFT orientation 10 s before landing. Payload orientation and vertical position were reconstructed from IMU quaternion and altimeter data, respectively. The XY-planar motion under 7 mph sustained wind speed was assumed to aid visualization of descent trajectory. The payload's reconstructed longitudinal vector was calibrated at the 5-deg launch rail.

As detailed in Section 2.4, the STEMCRaFT successfully jettisoned at 300 ft AGL after receiving RSO approval, descended under its own parachute, and landed with a KE of 16.5 lb-ft, meeting not only the 75 lb-ft threshold set by NASA Requirement 3.3 but also the 65 lb-ft bonus criteria (Table 6). Landing in an upright orientation is a VADL team-imposed requirement to maximize the effectiveness of RF transmission and was central in driving the design of the STEMCRaFT's passive, midair reorientation mechanism. To this end, using STEMCRaFT IMU quaternion and altimeter data, Figure 22 proves STEMCRaFT's upright landing orientation and stable descent, enabling the top antenna to position and fully extend in a most

favorable, vertically-upright orientation for RF transmission. Further, as highlighted in Section 5.2, the STEMCRaFT achieved a soft landing with a STEMnaut survivability index of 4.5, which is below the literature-informed threshold for the lowest possible injury risk. The soft landing is further discussed in the payload reflight certification in Section 4.5. VADL therefore assesses that the STEMCRaFT has achieved a fully nominal recovery, descent and landing at the 2025 USLI competition.

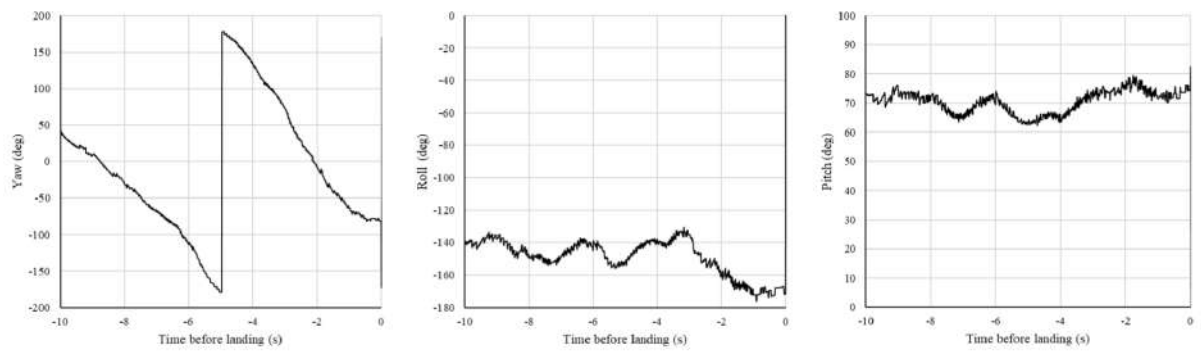


Figure 23: STEMCRaFT roll, pitch and yaw reconstructed from raw quaternion data 10 seconds before landing.

6 Project Management

6.1 Team Hours Spent

The team spent a total of 6721 hours on the project, broken down by project segment in Figure 24 below.

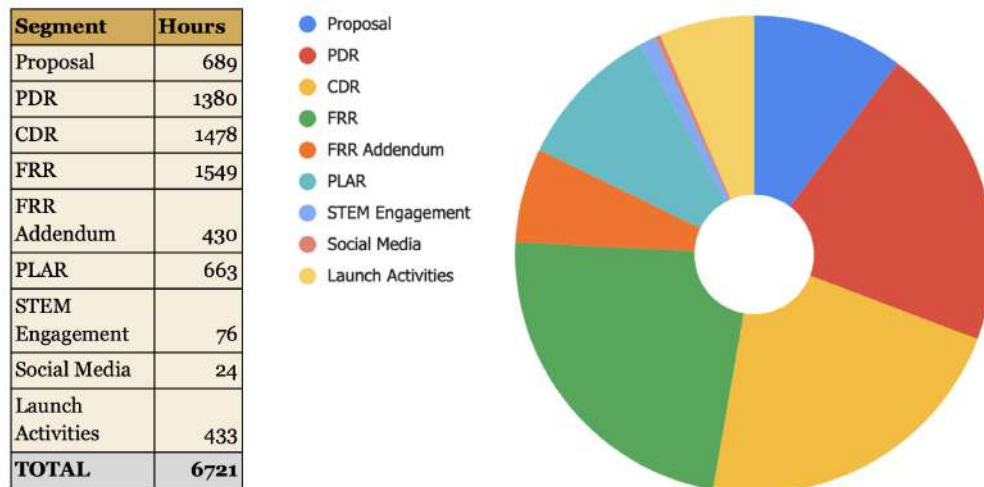


Figure 24: Total hours spent by project segment for the 2025 competition year.

6.2 STEM Engagement Summary

This project cycle, VADL focused on engaging the greater Nashville community through various educational activities. VADL wanted to ensure students were given access to aerospace engineering and followed the team on the USLI journey. One school, in particular, followed the team through PDR and CDR and

commemorated the end of the build season with the team at FRR—Millersville Elementary School, an underserved school in the greater Nashville area. Both the students and VADL greatly enjoyed collaborating throughout the project lifecycle. The team also explored new forms of delivering information about radiation and its effects on electronics through presentations, interactive tables, booths, and even a podcast! At the time of STEM Engagement Report submission, the team had reached a total of 1375 students, non-educators, parents, and other community members. Many of them interacted with the team through the parachute-making activity, which informed the community on the importance of parachute packaging and parachute design based on one of the many deterministic factors—mass. As detailed in the STEM Engagement Report, the community members thoroughly enjoyed the activities proposed by the team, as seen through the feedback received from students and their parents.

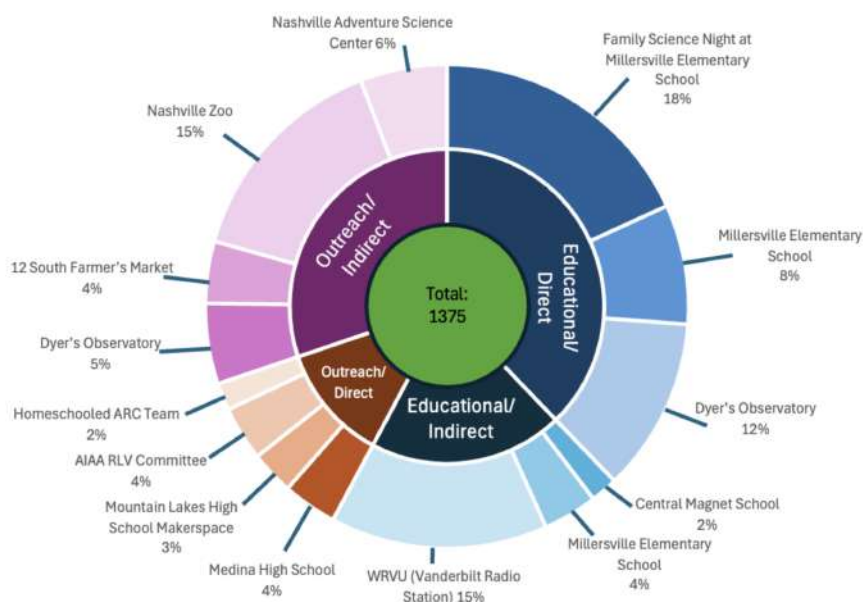


Figure 25: STEM engagement breakdown by events and categories.

6.3 Final Budget Summary

The team spent a total of \$13,516.20 for the 2024-2025 Student Launch competition. This amount is approximately 10% lower than the projected budget of \$14,956, but in line with the \$13,587 projection when excluding the contingency margin. The team managed to offset greater-than-expected payload costs – incurred through unanticipated parachute and antenna purchases – by utilizing existing inventory to reduce vehicle expenses. VADL will use the remainder of the \$16,000 funding allotment toward summer research projects.

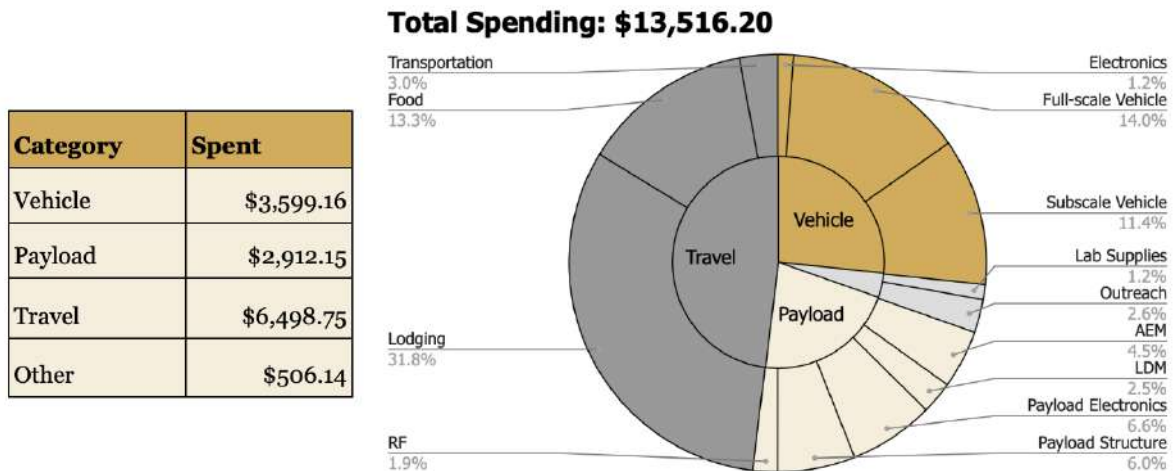


Figure 26: Final spending summary for the 2025 competition year.

6.4 Team Experience and Project Lessons Learned

The NASA Student Launch competition has provided VADL an invaluable experience through the full aerospace project lifecycle, from concept to completion. The team has learned numerous project management lessons along the way, some of which are summarized in Table 18 below.

Table 18: Key project management lessons learned and how they were addressed.

Lesson Learned	Mitigation Method
Cross-team collaboration is essential for ensuring everyone is on the same page.	In addition to nightly stand-up meetings, the team implemented breakout sessions between vehicle, payload, and electronics leads to discuss key developments from each subsystem in further detail.
Not all subsystems can always operate on parallel development timelines.	Staggered deadlines were implemented for finalizing work before milestone submissions – payload before vehicle, vehicle before simulations, etc.
The strongest way to drive design modifications and improvements is through integrated testing.	The team adopted a robust and rapid design, build, and test campaign, completing payload drop testing and demonstration flights nearly every weekend in the spring term.
Overly-defined roles and responsibilities can result in incomplete tasks.	The team agreed to adopt a mentality of selflessness and an “I can” approach to work, falling outside the regular responsibilities of team members to ensure that no tasks slip through the cracks.
Individuals work best and grow the most when given the opportunity to fail.	Team leadership shifted to a more delegation-prone mindset, enabling all team members to contribute. While sometimes resulting in minor setbacks in the short term, this approach produced the largest returns in the long term, resulting in the strongest team as a whole.

A Vehicle and Payload Reflight Certification

Vehicle and Recovery Inspection		
Component Type	Inspected Component	Status
Aerostructures	Nose Cone	No Damage
	LSM Bay	No Damage
	Drogue Bay	No Damage
	Avionics Bay	No Damage
	Main bay	No Damage
	Tail Section	No Damage
	Fins	No Damage
	Boat Tail	No Damage
	Motor Retention Tube	No Damage
Recovery System	18-inch Drogue Parachute	No Damage
	96-inch Main Parachute	No Damage
	Payload Parachute	No Damage
	Fire Blankets	No Damage
	Shock Cords	No Damage
	Fireballs	No Damage
	Quick Links	No Damage
	Lander Separation Mechanism	No Damage
	Eye Bolts	No Damage
	Blast Caps	No Damage
	Avionics Housing	No Damage
Coupler Sections	Nose Cone Retention Ring	No Damage
	Top Avionics Bulk Plate	No Damage
	Bottom Avionics Bulk Plate	No Damage

	Drogue Coupler	No Damage
	Main Coupler	No Damage

Vehicle and Recovery Inspection Methods		
Component Type	Verification	Results
Aerostructures	Verify nose cone sustained no damage	Verified
	Verify there are no structural damages, including buckling, in any of the airframe sections—LSM Bay, Recovery, and Tail.	Verified
	Verify there are no bulk plate radial holes stripped or have bearing damage	Verified
	Verify the fins have no structural damages and the fiberglass fillets are intact	Verified
	Verify the motor tube has no damages and the motor slides inside without any issues	Verified
	Verify the motor retention ring can screw back on the motor tube	Verified
	Verify the boat tail has no damages or shearing	Verified
Recovery System	Verify parachutes were not burned or ripped (including gores)	Verified
	Verify that fire blankets had no major burns or rips	Verified
	Verify that the parachute shroud lines incurred no damage	Verified
	Verify that eye bolts, quick links, and swivel links were not bent or compromised	Verified
	Verify that there were no tears in the shock cords	Verified
	Verify the avionics housing was undamaged and no fasteners were displaced	Verified

	Verify the blast caps stayed intact and incurred no damages	Verified
	Verify all the LSM bulk plates stayed intact and no damage was observed	Verified
	Verify the electromagnetic latches function properly—i.e. they latch and unlatch properly	Verified
Coupler Sections	Verify that the nose cone is retained safely and that no damage has been inflicted to the nose cone retention ring	Verified
	Verify the avionics bulk plates have no damage, buckling, or dents	Verified
	Verify all tapped and radial holes for stripping or damage	Verified
	Verify both main and drogue couplers for decoupling or shearing	Verified

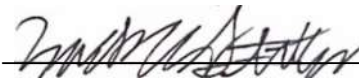
Signatures

Saksham Sharma, Vehicle Engineer:



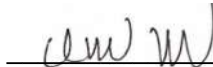
Date: 5/5/25

Zack Stutsman, Safety Officer:



Date: 5/5/25

Ian Morgan, Team Lead:



Date: 5/5/25

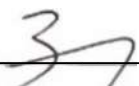
Payload Inspection	
Inspected Component	Status
Airframe	No Damage
Nosecone	No Damage
Bulk plate and Skeleton	No Damage
Electronics and RF System	No Damage
AEM	No Damage

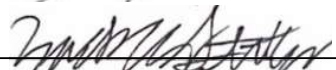
PDM	No Damage
LDM	Replaceable Shoes Deformed, Legs Undeformed

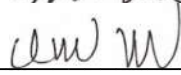
Payload Inspection Methods	
Verification	Result
Verify that the airframe and nose cone sustained no cracks or dents	Verified
Verify that nose cone retention ring sustained no cracks	Verified
Verify that all bulk plates were undeformed	Verified
Verify that structural aluminum rods sustained no bending	Verified
Verify that all 3D printed brackets and bays are undamaged	Verified
Verify that PDM latch still functions nominally and sustained no damage	Verified
Verify that all electrical connections are visibly intact and then verify that electronics can pass a functional check	Verified
Verify that shoes and/or legs deformed as intended	Verified
Verify that all bolts and heat inserts are intact and not damaged	Verified
Verify that STEMnauts were safely retained	Verified
Verify that batteries were not punctured	Verified
Verify that GPS was safely retained	Verified
Verify that AEM hardware was not damaged	Verified
Verify that AEM electrical connections are intact	Verified

Signatures

Rais Nurhidajat, Payload Engineer:  Date: 5/5/25

Patrick Chen, Electronics Engineer:  Date: 5/5/25

Zack Stutsman, Safety Officer:  Date: 5/5/25

Ian Morgan, Team Lead:  Date: 5/5/25

B References

- [1] Kenneth W. Finnegan. “Examining Ambiguities in the Automatic Packet Reporting System”. M.S. Thesis. San Luis Obispo, CA: California Polytechnic State University, San Luis Obispo, Dec. 2014.